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BATCH 15

Step 1: Aim

To write a menu-driven C program to perform different string operations on a student's full name using user-defined functions.

Step 2: Header Files

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
```

Step 3: Algorithm

1. Start the program.

2. Include the required header files.

3. Declare variables for the first name, middle name, last name, full name, and copied name.

4. Input the first name, middle name, and last name.

5. Combine all three names using `strcpy()` and `strcat()`.

6. Display the menu of operations.

7. Read the user's choice.

8. If the choice is 1, display the full name.

9. If the choice is 2, calculate the length using `strlen()`.

10. If the choice is 3, copy the full name using `strcpy()`.

11. If the choice is 4, compare the first and last names using `strcmp()`.

12. If the choice is 5, display the name in reverse order.

13. If the choice is 6, count vowels, consonants, digits, and spaces.

14. If the choice is 7, search for a word using `strstr()`.

15. If the choice is 8, exit the program.

16. If the choice is invalid, display an error message.

17. Repeat the menu until the user selects Exit.

18. Stop the program.

Step 4: C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(char f[], char m[], char l[], char full[]);
void findLength(char full[]);
void copyName(char source[], char dest[]);
void compareNames(char f[], char l[]);
```

```
void reverseNameOrder(char f[], char m[], char l[], char
rev[]);
void countCharacters(char full[]);
void searchWord(char full[], char word[]);

int main() {
    char f[50], m[50], l[50];
    char full[160], copied[160], rev[160], word[50];
    int choice;

    printf("Enter first name: ");
    scanf("%49s", f);

    printf("Enter middle name: ");
    scanf("%49s", m);

    printf("Enter last name: ");
    scanf("%49s", l);

    combineNames(f, m, l, full);

    do {
        printf("\n--- MENU ---\n");
        printf("1. Display full name\n");
        printf("2. Find length\n");
        printf("3. Copy full name\n");
        printf("4. Compare names\n");
        printf("5. Reverse name order\n");
        printf("6. Count characters\n");
        printf("7. Search a word\n");
        printf("8. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Full name: %s\n", full);
                break;

            case 2:
                findLength(full);
                break;

            case 3:
                copyName(full, copied);
```

```
                printf("Copied name: %s\n", copied);
                break;

            case 4:
                compareNames(f, l);
                break;

            case 5:
                reverseNameOrder(f, m, l, rev);
                printf("Reverse order: %s\n", rev);
                break;

            case 6:
                countCharacters(full);
                break;

            case 7:
                printf("Enter word to search: ");
                scanf("%49s", word);
                searchWord(full, word);
                break;

            case 8:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice!\n");
        }
    } while (choice != 8);

    return 0;
}

void combineNames(char f[], char m[], char l[], char full[])
{
    strcpy(full, f);
    strcat(full, " ");
    strcat(full, m);
    strcat(full, " ");
    strcat(full, l);
}

void findLength(char full[]) {
    printf("Length of full name: %zu\n", strlen(full));
```

```
}

void copyName(char source[], char dest[]) {
    strcpy(dest, source);
}

void compareNames(char f[], char l[]) {
    if (strcmp(f, l) == 0)
        printf("Names are equal.\n");
    else
        printf("Names are different.\n");
}

void reverseNameOrder(char f[], char m[], char l[], char
rev[]) {
    strcpy(rev, l);
    strcat(rev, " ");
    strcat(rev, m);
    strcat(rev, " ");
    strcat(rev, f);
}

void countCharacters(char full[]) {
    int i, v = 0, c = 0, d = 0, s = 0;
    char ch;

    for (i = 0; full[i] != '\0'; i++) {
        ch = tolower((unsigned char)full[i]);

        if (isalpha((unsigned char)ch)) {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
                v++;
            else
                c++;
        } else if (isdigit((unsigned char)ch)) {
            d++;
        } else if (isspace((unsigned char)ch)) {
            s++;
        }
    }

    printf("Vowels: %d\n", v);
    printf("Consonants: %d\n", c);
    printf("Digits: %d\n", d);
```

```
    printf("Spaces: %d\n", s);
}

void searchWord(char full[], char word[]) {
    if (strstr(full, word) != NULL)
        printf("Word found.\n");
    else
        printf("Word not found.\n");
}
```

Step 5: Sample Output

```
Enter first name: Rahul
Enter middle name: Kumar
Enter last name: Sharma

--- MENU ---
1. Display full name
2. Find length
3. Copy full name
4. Compare names
5. Reverse name order
6. Count characters
7. Search a word
8. Exit

Enter choice: 1
Full name: Rahul Kumar Sharma

Enter choice: 2
Length of full name: 18

Enter choice: 3
Copied name: Rahul Kumar Sharma

Enter choice: 4
Names are different.

Enter choice: 5
Reverse order: Sharma Kumar Rahul

Enter choice: 6
Vowels: 6
Consonants: 12
Digits: 0
Spaces: 2
```

```
Enter choice: 7
Enter word to search: Kumar
Word found.

Enter choice: 8
Exiting program.
```

Step 6: Result

The menu-driven C program was successfully implemented to perform string operations such as combining, finding length, copying, comparing, reversing, counting characters, and searching for a word using user-defined functions.